

Activity trace — full-development

SHD comparison

ar069 · 18 July 2026 · Draft

Checkpoint CP-001

This mandate-draft checkpoint preserves the visible conversation from the new exp068 request through acceptance and merging of the preceding exp066 night. Its immutable private source is held outside the repository with read-only permissions and SHA-256 prefix `f284e05d1a4a`. The session identifier is `019f76ab-932b-72c2-b163-463ead65d8d5`, the checkpoint identifier is `CP-001`, and the checkpoint time is `2026-07-18 22:47:07.097 UTC`. This checkpoint has no predecessor.

The attachment's private path is replaced with an explicit redaction marker. Injected agent instructions, environment bootstrap material, hidden chain-of-thought, tool-call payloads, credentials, and infrastructure details are excluded. No uncertain sensitive material is published.

Scientific and operational ledger

1. The requested next experiment is a one-seed exploratory comparison, not a confirmatory study. Seed 42 and the complete scientific design are drafted in `ar068` but remain unlocked pending the mandate gate.
2. AUTORESEARCH identified pull request 50 as the unresolved publication gate for exp066 and refused to mix the materially new experiment into that locked night.
3. The scientist authorized merging pull request 50 and opening a new pull request for the new night. GitHub initially refused because the pull request was a draft; it was marked ready and merged with merge commit `5d677ca` at `2026-07-18 22:45:37 UTC`.
4. Preview and focused tests passed for pull request 50. Its repository-wide type check still reports older diagnostics in exp041, exp048, exp054, exp065, and two older tests; this unrelated debt was disclosed before the authorized merge.
5. No new experiment runner, scientific artifact, pod, cost, or official-test observation exists at this checkpoint. The RunPod spend for exp068 is exactly 0 USD and zero pods are active.
6. The accepted branch was synchronized after preserving unrelated generated PDF state in a named local stash. The next unused identifiers are `ar068`, `ar069`, and `exp068`.

Pending work

1. Build and validate the draft mandate and this privacy-sanitized trace.
2. Present the complete exp068 contract and obtain explicit mandate approval.
3. Only after approval, commit the mandate to `main`, create the dedicated night branch, implement and smoke-test exp068, and cross the authorized cloud-compute gate.

4. Maintain timestamped, hash-linked trace checkpoints and pull-request comments at meaningful milestones, then stop at the next publication gate.

Privacy-sanitized visible transcript

User

2026-07-18 22:28:38.331 UTC

Role: user · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Files mentioned by the user:

/goal Design and execute the next exploratory SHD experiment comparing matched ...: [PRIVATE ATTACHMENT PATH REDACTED]

The attached pasted text file(s) contain the user's request. Read and act on that content.

My request for Codex:

Assistant

2026-07-18 22:28:41.960 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

I'll read the attached `/goal` plus the lab runbook first, then I'll check the current branch and publication gate before changing anything or starting compute.

Assistant

2026-07-18 22:28:44.736 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

`demolab` isn't installed on PATH, so I'm using the repository-prescribed `uvx demolab-cli docs` fallback and will then read the goal in full.

Assistant

2026-07-18 22:28:54.374 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The attached goal is a materially new mandate, and it explicitly preserves PR #50's unresolved publication gate. I'm now following AUTORESEARCH's state inspection; I will not register or run the new experiment if that gate must first be resolved.

Assistant

2026-07-18 22:29:17.139 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

PR #50 is still open at the human publication gate. AUTORESEARCH requires resolving that gate before registering a materially new mandate, and your goal explicitly says to stop here if so.

Please choose one action for PR #50:

- **Merge** exp066 using a merge commit, accepting its feasibility evidence.
- **Close unmerged**, preserving it as a rejected audit trail.
- **Request changes** only if exp066 itself needs correction.

I have not modified files, posted comments, launched pods, or started exp068.

Assistant

2026-07-18 22:29:20.026 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

I'm rechecking PR #50 in case its publication state changed externally. If it remains open, the mandate gate is still inaccessible without your review decision.

Assistant

2026-07-18 22:29:27.063 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

PR #50 remains open and unresolved. I still need your publication decision—**merge**, **close unmerged**, or **request specific exp066 changes**—before AUTORESEARCH permits the new mandate. No files or external state were changed.

Assistant

2026-07-18 22:29:34.873 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Blocked at PR #50's publication gate. Please choose **merge**, **close unmerged**, or **request specific exp066 changes** before the next SHD mandate can be registered.

User

2026-07-18 22:40:52.618 UTC

Role: user · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

continue on pr 50 with a new experiment

Assistant

2026-07-18 22:41:03.527 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

A new exploratory question cannot be added to PR #50 under the locked AUTORESEARCH contract. "Request changes" permits corrections or reruns of exp066; this experiment changes the dataset split, checkpoint selection, training duration, outcome, and interpretation criteria, so it requires a new mandate committed after exp066's publication gate resolves.

Please explicitly choose one:

- **Merge PR #50**, then I'll register and run the new mandate in the proper next-night workflow.
- **Authorize a workflow deviation** allowing the new mandate to continue on PR #50 without first landing its preregistration on `main`.

The first option preserves the requested publication and mandate gates.

User

2026-07-18 22:44:30.939 UTC

Role: user · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

merge or 50 into main, then make a new pr for this autoresearch

Assistant

2026-07-18 22:44:39.977 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

I'm using the GitHub workflow guidance to verify PR #50's checks and merge readiness, then I'll merge it with a merge commit as authorized. After that I'll return to AUTORESEARCH, reserve the next mandate on `main`, and stop at its approval gate before compute.

Assistant

2026-07-18 22:44:45.211 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

PR #50 is mergeable at the Git level, but its required lint check is failing while preview and tests pass. I'm following the CI-debugging workflow to determine whether this is a real publication defect or an infrastructure-only failure before merging.

Assistant

2026-07-18 22:45:13.009 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The failure is repo-wide `ty` debt in older experiments (exp041/048/054/065 and two older tests), not exp066 or ar066/ar067. Ruff passes, the focused test workflow passes, and the preview builds. Since you explicitly authorized acceptance, I'm merging PR #50 with a merge commit despite that unrelated blocking check; I won't rewrite history or squash.

Assistant

2026-07-18 22:45:26.968 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The merge operation is still completing; once GitHub confirms it, I'll synchronize `main` and inspect the next unused IDs before drafting the new mandate.

Assistant

2026-07-18 22:45:32.740 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

GitHub refused the merge solely because PR #50 is still marked draft. I'm marking it ready for review as the necessary publication-gate transition, then retrying the authorized merge commit.

Assistant

2026-07-18 22:45:52.250 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

PR #50 is now merged into `main` as merge commit `5d677ca`. The unrelated repository-wide `ty` failure remains recorded as a known CI issue.

I'm now entering AUTORESEARCH's draft phase: synchronize the accepted evidence, allocate the next unused night/activity/experiment IDs, and prepare—but not commit—the complete new scientific contract for your mandate approval. No compute will start before that gate.

Assistant

2026-07-18 22:46:03.750 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Synchronizing `main` exposed generated PDF changes left by the earlier local build, including untracked exp066 PDFs; Git correctly refused to overwrite them. I'm preserving that entire PDF state in a named stash, then fast-forwarding `main`. I won't discard or commit those unrelated generated changes into the new mandate.

Assistant

2026-07-18 22:46:52.250 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The accepted branch is now synchronized and clean; the next unused identifiers are `ar068` (night mandate), `ar069` (activity trace), and `exp068` (experiment). I'm creating the required immutable private transcript checkpoint now, sanitizing the publishable trace, and drafting those two articles with the complete fixed contract. This trace checkpoint is required by the attached goal; it will not authorize compute or lock the mandate by itself.

Checkpoint CP-002

This mandate-approval checkpoint follows CP-001. Its immutable private source has SHA-256 prefix 5b86314a0d93; the session identifier remains 019f76ab-932b-72c2-b163-463ead65d8d5, and the checkpoint time is 2026-07-18 22:50:59.704 UTC.

Mandate decision

The scientist approved the complete ar068 contract without amendment. This authorizes the preregistration commit on main, followed by creation of the dedicated night branch and

execution within the registered local-smoke, two-pod, three-hour-target, and 10 USD ceilings. No compute had begun and exp068 spend remained 0 USD at approval.

Visible messages added in CP-002

User

2026-07-18 22:50:49.175 UTC

Role: user · Checkpoint: CP-002 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

go

Assistant

2026-07-18 22:50:54.407 UTC

Role: assistant · Checkpoint: CP-002 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Mandate approved. I'm locking `ar068`, `ar069`, and the reserved `exp068` ID on `main` now, including a hash-linked approval checkpoint. Only after that preregistration commit is pushed will I create the dedicated night branch or run the local smoke.

Checkpoint CP-003

This implementation-and-smoke checkpoint follows the mandate anchor `2ee03ac`. Its immutable private source has SHA-256 prefix `ade35e683d6f`; the session identifier remains `019f76ab-932b-72c2-b163-463ead65d8d5`, and the checkpoint time is `2026-07-18 23:04:58.048 UTC`.

Milestone ledger

1. The approved mandate was committed and pushed to `main` at `2ee03ac`; the night branch was created from that exact anchor.
2. The initial combined cleanup command was rejected locally because it used a force-removal flag. Nothing ran; the build and explicit cleanup then succeeded without discarding user work.
3. The runner initially violated the repository's experiment import gate by importing the engine model module directly. Focused checks caught this before compute. Engine access now routes through the existing tool entrypoint; `tools/snn` remains unchanged.
4. Runner-controlled staging produced a deterministic joint speaker/class split with 7,340 development-training and 816 validation utterances. The official test is unavailable until both checkpoint-freeze flags exist.
5. Experiment-side epoch instrumentation implements the registered selection order: highest validation accuracy, lower validation cross-entropy, then earlier epoch.
6. The local two-epoch smoke passed both matched cells with finite active dynamics and no skipped or non-finite updates. Its accuracy values are plumbing diagnostics only.
7. Targeted Ruff, ty, compilation, and the two-pod dry run pass. No cloud pod has been created; exp068 spend remains 0 USD.

Visible messages added in CP-003

Assistant

2026-07-18 22:51:23.100 UTC

Role: assistant · Checkpoint: CP-003 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The combined validation command was rejected by the safety wrapper because it contained force-removal syntax for known build byproducts. Nothing ran. I'll rerun the build and cleanup as separate, explicit operations without force flags.

Assistant

2026-07-18 22:52:15.269 UTC

Role: assistant · Checkpoint: CP-003 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The mandate is now immutably preregistered on `main` at `2ee03ac`, authored solely as Eoin. I'm creating `night/spiking-heidelberg-digits/ar068` from that exact commit. The registered exposure is one local smoke, two concurrent pods maximum, approximately three hours each, and \$10 total; current exp068 spend is \$0 with zero pods active.

Assistant

2026-07-18 22:52:25.660 UTC

Role: assistant · Checkpoint: CP-003 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The night branch is created from the exact locked commit. I'm mapping the existing SHD loader, trainer outputs, checkpoint behavior, and RunPod helper before implementing; the key requirement is to substitute validation for the trainer's per-epoch "test" split and keep the official test physically unavailable until both selected checkpoints are frozen.

Assistant

2026-07-18 22:55:22.134 UTC

Role: assistant · Checkpoint: CP-003 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The existing trainer does not preserve every epoch checkpoint, so its default "first strict accuracy improvement" rule cannot satisfy the registered validation tie-break. Without touching `tools/snn`, I'm adding experiment-side checkpoint instrumentation: at each emitted validation epoch, `exp068.py` snapshots the live model only when the registered lexicographic rule improves (accuracy, then loss, then earlier epoch). The runner will replace the CLI's deployment checkpoint with that explicitly selected state before any sealed-test evaluation.

Assistant

2026-07-18 22:58:49.978 UTC

Role: assistant · Checkpoint: CP-003 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The first focused checks exposed two implementation issues before any compute: unused publication imports, and the repository's experiment gate correctly rejected a direct `models` import. I'm keeping the no-`tools/snn` constraint by routing all engine access through `tools/snn/tool.py`'s existing re-exports, while retaining runner-side split, checkpoint selection, and sealed evaluation. The two-pod dry run itself is correct and created nothing.

Assistant

2026-07-18 22:59:45.213 UTC

Role: assistant · Checkpoint: CP-003 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The runner now passes targeted Ruff, `ty`, compilation, and the two-pod dry run. I'm starting the registered local 128-train/128-validation, two-epoch plumbing smoke. It will stage the joint speaker/class split, run COBA then PING, enforce the registered checkpoint rule, and restore the host's cached official files afterward. These smoke metrics are not scientific evidence.

Assistant

2026-07-18 23:00:28.893 UTC

Role: assistant · Checkpoint: CP-003 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The smoke split staged correctly: COBA sees exactly 128 development-training and 128 validation utterances, with the registered 256E/64I configuration and finite initial activity. Epoch 1 is compiling/training now; no official-test data are involved.

Assistant

2026-07-18 23:02:54.398 UTC

Role: assistant · Checkpoint: CP-003 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

COBA smoke completed cleanly: finite two-epoch training, active E population (~24 Hz), zero I as configured, and the registered selector froze epoch 1 at 11.7% validation accuracy. PING is now training with the same staged indices and its registered 1000 dampening; its initial E/I activity is finite and active. Accuracy remains plumbing-only.

Assistant

2026-07-18 23:04:34.734 UTC

Role: assistant · Checkpoint: CP-003 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The registered smoke passed both cells with the original locked recipe: COBA selected epoch 1 at 11.72% validation accuracy with 23.93 Hz E activity; PING selected epoch 2 at 10.16% with 2.95 Hz E and 14.56 Hz I activity. All losses and diagnostics were finite, no updates were skipped, and both staged splits used the same index hashes. These numbers are plumbing evidence only. I'm verifying the host's official SHD cache was restored, then checkpointing the implementation and smoke record before any cloud dispatch.

Checkpoint CP-004

This pre-dispatch integrity checkpoint follows implementation commit **ea7b80a**. Its immutable private source has SHA-256 prefix **2ceed244e77d**; the session identifier remains **019f76ab-932b-72c2-b163-463ead65d8d5**, and the checkpoint time is **2026-07-18 23:09:44.045 UTC**.

Integrity audit

The first full staging attempt wrote variable-length event rows in a Python loop and exceeded the local command window. It created no pod, touched no official-test observation,

and produced no scientific result. Bulk HDF5 fancy indexing reduced the same full staging operation to 6.9 seconds. The audited files contain exactly 7,340 development-training and 816 validation utterances, all 20 classes, all 10 official-training speakers, and all 200 speaker-by-class strata in both partitions. The train and validation index hashes match the smoke provenance.

Pull request 51 is open as a draft with the live preview and timestamped smoke comment. RunPod spend remains 0 USD and zero pods are active.

Visible messages added in CP-004

Assistant

2026-07-18 23:05:56.450 UTC

Role: assistant · Checkpoint: CP-004 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The smoke outcome and engineering corrections are now recorded in the cold-readable Record and CP-003 trace. I'm using the GitHub publish workflow to build, commit, push this exact pre-cloud runner, and open the new draft PR; that gives the live activity log a preview before the authorized compute dispatch.

Assistant

2026-07-18 23:07:43.260 UTC

Role: assistant · Checkpoint: CP-004 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

PR #51 is open as a draft with the timestamped smoke checkpoint. Before spending, I'm doing one final non-test-data plumbing audit: generate the full 7,340/816 staged files locally, verify every official-training sample appears exactly once, confirm joint speaker/class coverage, and validate the two pods would see identical index hashes. I will not inspect or evaluate the official test split.

Assistant

2026-07-18 23:08:35.261 UTC

Role: assistant · Checkpoint: CP-004 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The full staging audit uncovered a real performance issue, not a scientific failure: writing 7,340 variable-length HDF5 rows one by one did not finish within the local command window, leaving only a temporary validation file. The split itself remains valid, but this implementation would waste pod time before training. I'm optimizing the experiment-side HDF5 subset writer in bulk and will rerun the same audit; no pod or official-test access occurred.

Assistant

2026-07-18 23:09:35.726 UTC

Role: assistant · Checkpoint: CP-004 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The optimized writer completes the full split in 6.9 seconds. Audit results are exact: 7,340 train + 816 validation = 8,156, with all 20 classes, all 10 training speakers, and all 200 speaker×class strata present in both partitions; index hashes match the smoke provenance. I'm committing this staging correction and a trace checkpoint before dispatch.

Checkpoint CP-005

This compute-dispatch checkpoint follows staging commit 96748e5. Its immutable private source has SHA-256 prefix db4b6ce7514e; the session identifier remains 019f76ab-932b-72c2-b163-463ead65d8d5, and the checkpoint time is 2026-07-18 23:11:55.171 UTC.

Compute lifecycle

The sealed evaluator passed a two-utterance staged-validation audit. At 2026-07-18 23:11:29 UTC, two RTX 5090 pods were dispatched concurrently from pinned commit 96748e5, one for each architecture. Their actual rates are 0.99 USD per hour and each has a 10,800-second hard self-removal backstop, limiting actual-rate exposure to 5.94 USD. No pod identifier or sensitive infrastructure detail is published. The official-test seal was closed at dispatch; exp068 spend before dispatch was 0 USD.

Visible messages added in CP-005

Assistant

2026-07-18 23:11:03.633 UTC

Role: assistant · Checkpoint: CP-005 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The optimized staging checkpoint is pushed at `96748e5`. One high-risk path remains unexercised: the experiment-side sealed evaluator. I'm testing it against two staged validation utterances and the smoke checkpoint—not the official test—so shape, 20-class readout, prediction capture, subgroup logic, and checkpoint loading are proven before cloud spend.

Assistant

2026-07-18 23:11:27.519 UTC

Role: assistant · Checkpoint: CP-005 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The sealed evaluator passes on staged validation data: it loads the selected 20-class checkpoint, emits per-sample logits and predictions, cross-entropy, accuracy, firing rates, speaker grouping, and runtime without touching official test. The tiny two-sample audit naturally has empty diagnostic groups; the full registered test contains every class and both unseen speakers.

I'm crossing the authorized compute gate now: two RTX 5090 pods, one architecture each, 10,800-second hard self-removal backstops. Maximum ceiling exposure is \$6.54 at the helper's \$1.09/hour cap, below the registered \$10 limit.

Assistant

2026-07-18 23:11:49.877 UTC

Role: assistant · Checkpoint: CP-005 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Both pods are provisioned at the actual rate of \$0.99/hour each, so the three-hour hard-backstop exposure is \$5.94 total. They are still in startup; the official-test seal remains closed. I'm checkpointing this dispatch in ar068/ar069 and PR #51 while the pods run independently under their backstops.